Mini Neural Network Classifier

Machine Learning Assignment 14

October 7, 2026

1 Introduction

In this assignment, we implement a simple Neural Network (using Scikit-Learn's `MLPClassifier` with a single hidden layer of 100 neurons) and compare its performance against a much simpler baseline model: Logistic Regression. We train both models on the high-dimensional Breast Cancer dataset. Before training, the data was rigorously standardized, as neural networks are notoriously sensitive to unscaled input features.

2 Performance Comparison

The accuracy of both models on the test set is reported below:

Model	Accuracy
Logistic Regression (Baseline)	0.9825
Neural Network (MLP)	0.9825

Table 1: Model Accuracy Comparison.

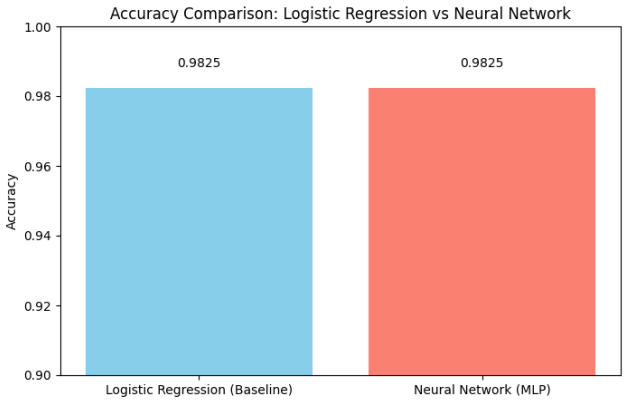


Figure 1: Accuracy comparison between Logistic Regression and the Neural Network.

Interestingly, both models achieved the exact same high accuracy (98.25%) on the test set. This tells us that the boundary separating malignant from benign tumors in this specific dataset is largely linear. When a problem is inherently linear, injecting massive complexity (via a Neural Network) does not yield better predictive power, as the simpler baseline is already capturing the optimal decision boundary.

3 Discussion: Neural Networks vs. Simpler Models

Advantage of the Neural Network

The primary advantage of a Neural Network is its **ability to model highly complex, non-linear relationships**. While Logistic Regression assumes a straight-line boundary between classes, a Neural Network can mathematically twist, bend, and warp its decision boundary to fit incredibly convoluted patterns (e.g., recognizing faces in pixels or translating languages). If the dataset had hidden non-linear interactions between features, the Neural Network would likely have vastly outperformed the linear baseline.

Disadvantage of the Neural Network

A major disadvantage of Neural Networks compared to Logistic Regression is **a severe lack of interpretability (the "Black Box" problem)**. In Logistic Regression, we can look exactly at the learned weights (coefficients) to confidently say, "Feature X directly increases the odds of cancer by $Y\%$." In a Neural Network, the input features are multiplied across thousands of hidden weights and non-linear activation functions. It becomes virtually impossible to trace exactly *why* the network made a specific prediction, which is a major hurdle in regulated industries like medicine or finance. Additionally, Neural Networks require vastly more computational resources and meticulous hyperparameter tuning to train successfully.